LP Duality, Complementary Slackness, and Theorems of the Alternative

A study guide for the material of Homework 3

Optimization (Caramanis & Sanghavi) — companion notes

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1 Why duality

Every linear program comes with a twin, its *dual*. The dual answers a question the primal cannot answer by itself: *how do I know a proposed solution is optimal?* A feasible x proves "the optimum is at most $c^\mathsf{T}x$"; a dual-feasible p proves "the optimum is at least $p^\mathsf{T}b$." When the two numbers meet, optimality is *certified*. The same mechanism certifies infeasibility (Farkas' lemma) and explains which constraints matter at the optimum (complementary slackness). Homework 3 exercises the three core skills:

- mechanically forming a dual, with every sign right (Problem 1);
- converting a logical statement about linear inequalities into an LP and reading off the alternative from its dual (Problem 2);
- exploiting the *structure* of dual solutions, namely sparsity of vertices and complementary slackness (Problem 3).

2 Forming the dual

2.1 The dictionary

For a **minimisation** primal with cost c, constraint matrix A (rows a_i^T, columns A_j) and right-hand side b:

Primal: min $c^\mathsf{T}x$		**Dual:** max $p^\mathsf{T}b$	
i-th constraint	$a_i^\mathsf{T}x \geq b_i$	i-th variable	$p_i \geq 0$
	$a_i^\mathsf{T}x \leq b_i$		$p_i \leq 0$
	$a_i^\mathsf{T}x = b_i$		p_i free
j-th variable	$x_j \geq 0$	j-th constraint	$p^\mathsf{T}A_j \leq c_j$
	$x_j \leq 0$		$p^\mathsf{T}A_j \geq c_j$
	x_j free		$p^\mathsf{T}A_j = c_j$

One dual variable per primal constraint; one dual constraint per primal variable. For a **maximisation** primal, read the table right-to-left.

Mnemonic ("natural ↔ nonnegative"). In a min problem the natural constraint direction is $\geq$; in a max problem it is $\leq$. A natural constraint gives a nonnegative multiplier, an unnatural one a nonpositive multiplier, an equality a free one. Symmetrically, a nonnegative variable gives a natural dual constraint, a nonpositive variable an unnatural one, a free variable an equality.

2.2 Where the dictionary comes from: the Lagrangian

Relax the general constraints with prices p and define

$$g(p) = \min_{x \text{ obeying its sign constraints}} \left[c^\mathsf{T}x + p^\mathsf{T}(b - Ax)\right] = p^\mathsf{T}b + \min_x \sum_j (c_j - p^\mathsf{T}A_j)x_j.$$

1. **Signs of** p are chosen so that the penalty $p^\mathsf{T}(b - Ax)$ is ≤ 0 at every feasible x. Then $g(p) \leq c^\mathsf{T}x$ for all feasible x: every such p yields a lower bound.
2. **Dual constraints** are exactly the conditions for the inner minimum to be finite (it is then 0). If $x_j \geq 0$ we need $c_j - p^\mathsf{T}A_j \geq 0$; if $x_j \leq 0$, ≤ 0; if x_j is free, $= 0$.
3. The dual is "find the best lower bound": $\max_p g(p)$.

If you forget the table, this three-line derivation rebuilds it, and it is the same construction that defines the dual of a general convex program later in the course.

2.3 The standard pairs worth memorising

Primal	Dual
$\min c^\mathsf{T}x$ s.t. $Ax = b,\ x \geq 0$ (standard form)	$\max p^\mathsf{T}b$ s.t. $A^\mathsf{T}p \leq c$
$\min c^\mathsf{T}x$ s.t. $Ax \geq b,\ x \geq 0$ (symmetric form)	$\max p^\mathsf{T}b$ s.t. $A^\mathsf{T}p \leq c,\ p \geq 0$
$\min c^\mathsf{T}x$ s.t. $Ax \geq b$	$\max p^\mathsf{T}b$ s.t. $A^\mathsf{T}p = c,\ p \geq 0$
$\max c^\mathsf{T}x$ s.t. $Ax \leq b,\ x \geq 0$	$\min b^\mathsf{T}y$ s.t. $A^\mathsf{T}y \geq c,\ y \geq 0$

Example 2.1 (HW3 Problem 1). $\min x_1 - x_2$ with rows of type $\leq$, $\geq$, $=$ and variables $x_1 \leq 0$, $x_2, x_3 \geq 0$, x_4 free. Rows give $p_1 \leq 0$, $p_2 \geq 0$, p_3 free. Columns give, in order, a "$\geq c_1$" constraint, two "$\leq c_j$" constraints and one equality. The objective is $b^\mathsf{T} p = 3p_2 + 6p_3$. For this instance the dual is infeasible and the primal unbounded, a reminder that "compute the dual" and "the dual has a solution" are different questions.

Common pitfall.

- **A variable with no stated sign is free**, and produces an *equality* in the dual. (In HW3 P1, x_4.)
- **The dictionary depends on min vs. max.** Applying the min-table to a max problem flips every sign. When unsure, convert to a min problem first, or redo the Lagrangian argument.
- **Do not convert to standard form first** unless asked: it works (equivalent primals have equivalent duals) but doubles the bookkeeping and the chance of error.
- **Check dimensions**: $p \in \mathbb{R}^{\#\text{constraints}}$, and there are #variables dual constraints. Sign restrictions on x are *not* counted as constraints here.

Proposition 2.2. *The dual of the dual is the primal. Equivalent formulations of the primal (slack variables, splitting $x = x^+ - x^-$, negating rows) have equivalent duals.*

3 Weak and strong duality

Theorem 3.1 (Weak duality)**.** *If x is primal feasible (min problem) and p is dual feasible, then $p^\mathsf{T} b \leq c^\mathsf{T} x$.*

Proof. $c^\mathsf{T} x - p^\mathsf{T} b = \sum_j (c_j - p^\mathsf{T} A_j) x_j + \sum_i p_i (a_i^\mathsf{T} x - b_i)$, and the dictionary was built so that every term in both sums is ≥ 0. □



Immediate consequences: (a) primal unbounded ⇒ dual infeasible; (b) dual unbounded ⇒ primal infeasible; (c) if $c^\mathsf{T} x = p^\mathsf{T} b$ for a feasible pair, both are optimal.

Theorem 3.2 (Strong duality)**.** *If an LP has an optimal solution, so does its dual, and the optimal values are equal.*

For LPs there is no constraint qualification and no duality gap: feasibility plus boundedness is enough. (For general convex programs one needs a condition such as Slater's.)

The possibilities table.

	Dual: finite optimum	Dual: unbounded	Dual: infeasible
Primal: finite optimum	**possible** (equal values)	impossible	impossible
Primal: unbounded	impossible	impossible	**possible**
Primal: infeasible	impossible	**possible**	**possible**

HW3 P1 lives in the middle-right cell. The bottom-right cell is real: $\min x_1 + 2x_2$ s.t. $x_1 + x_2 = 1$, $2x_1 + 2x_2 = 3$ is infeasible, and its dual $\max p_1 + 3p_2$ s.t. $p_1 + 2p_2 = 1$, $p_1 + 2p_2 = 2$ is infeasible too.

Common pitfall. "Dual infeasible" does *not* imply "primal unbounded": the primal might be infeasible. To conclude unboundedness you must also exhibit a primal feasible point (as in P1: $x^0 = (0, 0, 2, 2)$).

Interpretation. The optimal p_i is the *shadow price* of constraint i: the rate of change of the optimal cost per unit change of b_i (when the optimal basis is nondegenerate). A constraint with slack has price zero, which is the content of the next section.

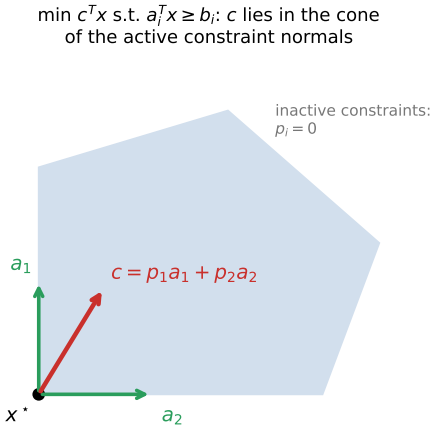
4 Complementary slackness

Theorem 4.1 (Complementary slackness)**.** *Let x be primal feasible and p dual feasible. They are both optimal if and only if*

$$p_i \,(a_i^{\mathsf{T}} x - b_i) = 0 \quad \forall i \qquad \textit{and} \qquad (c_j - p^{\mathsf{T}} A_j)\, x_j = 0 \quad \forall j.$$

Proof. From the proof of weak duality, $c^{\mathsf{T}} x - p^{\mathsf{T}} b$ is a sum of nonnegative terms, namely exactly the products above. The gap is zero iff every term is zero, and a zero gap is equivalent to optimality of both (weak + strong duality). □

In words: *a positive price forces a tight constraint; a slack constraint forces a zero price* (and likewise for variables versus dual constraints).



Geometry. For $\min c^\mathsf{T}x$ s.t. $a_i^\mathsf{T}x \geq b_i$, the dual constraint is $c = \sum_i p_i a_i$ with $p \geq 0$, and complementary slackness allows $p_i > 0$ only on active constraints. So $x^\star$ is optimal iff c lies in the cone generated by the normals of the constraints active at $x^\star$. This is the LP case of the KKT conditions.

Recipe. Using complementary slackness to certify or find solutions.

1. Given a candidate x: check primal feasibility; list the tight constraints and the nonzero variables.
2. Set $p_i = 0$ for every slack constraint; impose $p^\mathsf{T}A_j = c_j$ for every j with $x_j \neq 0$. Solve this (usually square) linear system.
3. Check that the resulting p is dual feasible (signs and remaining constraints). If yes, x and p are optimal; if the system is inconsistent or p infeasible (and x is nondegenerate), x is not optimal.

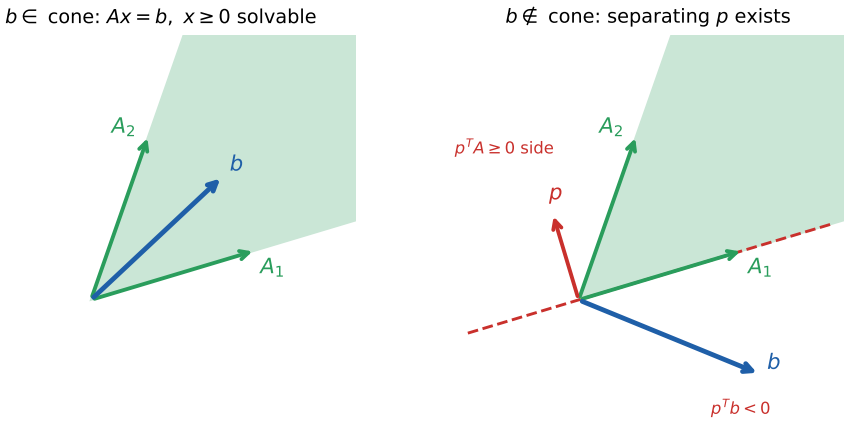
Example 4.2. $\min 13x_1 + 10x_2 + 6x_3$ s.t. $5x_1 + x_2 + 3x_3 = 8$, $3x_1 + x_2 = 3$, $x \geq 0$, candidate $x = (1, 0, 1)$. Dual: $\max 8p_1 + 3p_2$ s.t. $5p_1 + 3p_2 \leq 13$, $p_1 + p_2 \leq 10$, $3p_1 \leq 6$. Since $x_1, x_3 > 0$: $5p_1 + 3p_2 = 13$ and $3p_1 = 6$, so $p = (2, 1)$. Check the remaining constraint: $2 + 1 \leq 10$. Feasible, so both are optimal; values $13 + 6 = 19 = 16 + 3$.

5 Theorems of the alternative

A *theorem of the alternative* states that exactly one of two linear systems is solvable. The second system is a **certificate**: a short proof that the first has no solution.

Theorem 5.1 (Farkas' lemma)**.** *Exactly one of the following holds: (a) $\exists x \geq 0$ with $Ax = b$; (b) $\exists p$ with $p^\mathsf{T}A \geq 0$ and $p^\mathsf{T}b < 0$.*

Proof. Not both: $0 \leq (p^\mathsf{T}A)x = p^\mathsf{T}b < 0$ is absurd. *At least one:* consider $\max 0^\mathsf{T}x$ s.t. $Ax = b$, $x \geq 0$, whose dual is $\min p^\mathsf{T}b$ s.t. $p^\mathsf{T}A \geq 0$. The dual is feasible ($p = 0$). If (a) fails the primal is infeasible, so the dual cannot have a finite optimum (strong duality would hand the primal an optimal solution); hence the dual is unbounded, giving p with $p^\mathsf{T}A \geq 0$, $p^\mathsf{T}b < 0$. □



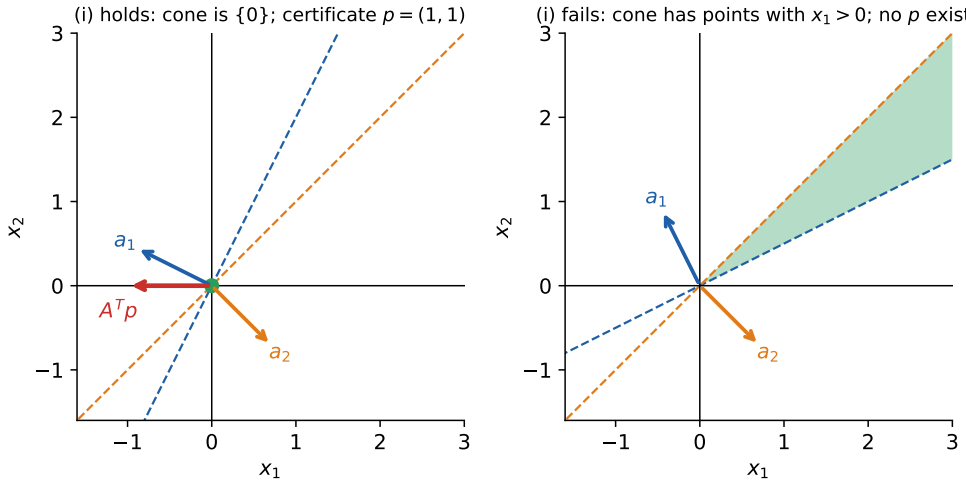
Geometry. (a) says b lies in the cone generated by the columns of A. If it does not, a hyperplane through the origin separates b from that (closed, convex) cone, and its normal is p. Farkas' lemma is the separating hyperplane theorem for finitely generated cones.

The family. All are proved the same way.

System I	System II (certificate that I fails)	Name
$Ax = b,\ x \geq 0$	$p^\mathsf{T}A \geq 0,\ p^\mathsf{T}b < 0$	Farkas
$Ax \leq b$	$p \geq 0,\ A^\mathsf{T}p = 0,\ b^\mathsf{T}p < 0$	Farkas (inequality form)
$Ax \leq b,\ x \geq 0$	$p \geq 0,\ A^\mathsf{T}p \geq 0,\ b^\mathsf{T}p < 0$	Farkas (mixed)
$Ax < 0$	$p \geq 0,\ p \neq 0,\ A^\mathsf{T}p = 0$	Gordan
$Ax \geq 0,\ x \geq 0,\ x_1 > 0$	$p \geq 0,\ p^\mathsf{T}A \leq 0,\ p^\mathsf{T}A_1 < 0$	HW3 Problem 2

Recipe. Proving a theorem of the alternative.

1. *Easy direction (not both).* Assume both systems hold, multiply the inequalities of I by the multipliers of II and add. You should reach $0 \leq \cdots < 0$ or "a sum of nonpositive terms is ≥ 0, so each term vanishes."
2. *Hard direction.* Encode system I as an LP: objective 0 for a pure feasibility question, or the strict quantity as the objective (HW3 P2: $\min -x_1$ over the cone). Make sure the LP is *feasible*.
3. Take the dual with the dictionary.
4. Conclude with strong duality: "primal has an optimal solution $\Rightarrow$ dual feasible," or "primal infeasible and dual feasible $\Rightarrow$ dual unbounded." The dual-feasible (or improving) vector is the certificate.
5. *Strict inequalities.* LPs cannot express them. If the rest of the system is homogeneous, replace $p^\mathsf{T}A_1 < 0$ by $p^\mathsf{T}A_1 \leq -1$ and note that rescaling p makes these equivalent.



The two-panel figure shows HW3 P2 in $\mathbb{R}^2$. Left: the cone $\{x \geq 0, Ax \geq 0\}$ is $\{0\}$, and the certificate $A^\mathsf{T} p = (-1, 0)$ is a nonnegative combination of the rows that is ≤ 0 with negative first entry, so "$(A^\mathsf{T} p)^\mathsf{T} x \geq 0$" can hold on the orthant only when $x_1 = 0$. Right: the cone contains points with $x_1 > 0$, so no certificate exists.

Common pitfall. The implication "$Ax \geq 0,\ x \geq 0 \Rightarrow x_1 = 0$" is a statement about *all* x; its proof via duality needs the LP to be feasible (here $x = 0$ works) so that strong duality applies. Always state why the LP you wrote is feasible and bounded before invoking strong duality.

6 Vertices are sparse: basic solutions, Carathéodory, Helly

Definition 6.1. For a standard-form polyhedron $P = \{x \in \mathbb{R}^N : Mx = d,\ x \geq 0\}$ with $M \in \mathbb{R}^{k \times N}$, a *basic feasible solution* (BFS) is a feasible x whose positive entries correspond to linearly independent columns of M. BFS = vertex = extreme point.

Theorem 6.2.

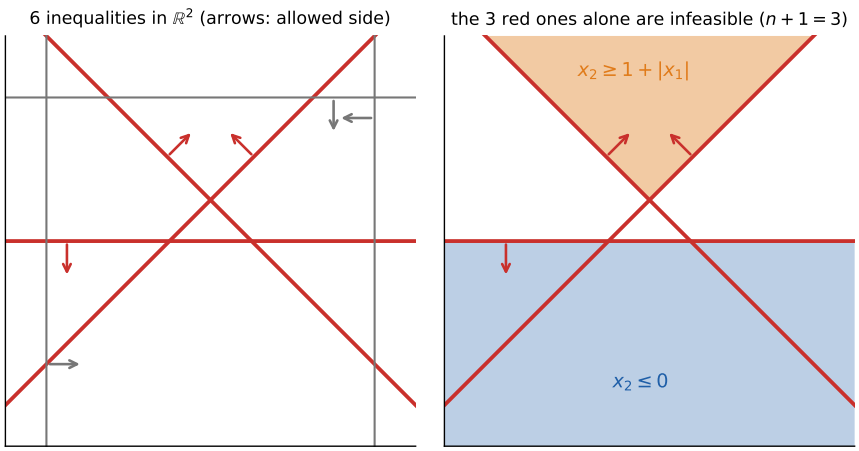
1. *A BFS has at most* $\operatorname{rank} M \leq k$ *nonzero entries.*
2. *If $P \neq \emptyset$ it has a BFS. If the LP $\min c^\mathsf{T} x$ over P has an optimal solution, it has an optimal BFS.*

Proof idea for existence. If the columns on the support of x are dependent, take a null vector z supported there, and move along $x - \theta z$ until a positive entry hits zero. Equalities are preserved, nonnegativity is preserved, the support shrinks. For the optimal version, choose the sign of z so the cost does not increase. Repeat. □

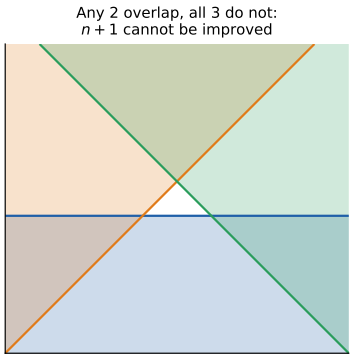
Key idea. Count equality constraints to bound nonzeros. Whenever an object is described by "k linear equations in nonnegative unknowns," some solution uses at most k of the unknowns. Three classical faces of this one fact:

- *Carathéodory:* a point in the convex hull of a set in $\mathbb{R}^n$ is a convex combination of at most $n + 1$ of its points (n equations for the point, 1 for $\sum \lambda_i = 1$).
- *HW3 P3 / Helly:* an infeasible system $Ax \leq b$ in n variables has an infeasible subsystem of at most $n + 1$ inequalities (n equations $A^\mathsf{T} p = 0$, 1 normalisation).
- *Sparse LP solutions:* an LP with k constraints needs at most k nonzero variables.

The argument of HW3 P3 in four lines. (1) $\min t$ s.t. $a_i^\mathsf{T} x - t \leq b_i$ has optimum $t^\star > 0$ iff $Ax \leq b$ is infeasible. (2) Its dual is $\max -b^\mathsf{T} p$ s.t. $A^\mathsf{T} p = 0,\ \mathbf{1}^\mathsf{T} p = 1,\ p \geq 0$: standard form, $n + 1$ equalities. (3) Take an optimal BFS $p^\star$, support S, $|S| \leq n + 1$. (4) Restricted to S, the pair $(x^\star, t^\star)$, $p^\star_S$ is feasible and complementary, hence optimal for the subsystem, whose optimum is therefore $t^\star > 0$: infeasible.



Theorem 6.3 (Helly, for half-spaces)**.** *If every $n+1$ members of a finite family of half-spaces in $\mathbb{R}^n$ have a common point, the whole family has a common point. (True for arbitrary convex sets as well.)*



The number $n+1$ is best possible: $x_j \geq 1$ $(j \leq n)$ together with $\sum_j x_j \leq n-1$ are jointly infeasible, yet every n of them are feasible.

Common pitfall. The bound counts *equality rows of the dual*, not primal constraints. If the dual has redundant rows the true bound is the rank, which can be smaller, never larger. Also, "at most $n+1$" says nothing about *which* subsystem: different vertices of the certificate polyhedron give different infeasible subsystems.

Practical note. When testing feasibility numerically, solve the phase-1 problem $\min t$ rather than passing a zero objective and trusting a status flag: the phase-1 LP is always feasible, its optimum quantifies the violation, and its dual solution *is* the Farkas certificate.

7 Self-test questions

Self-test 1. Write the dual of $\max\ 2x_1 + x_2$ s.t. $x_1 + x_2 \leq 4$, $x_1 - x_2 \geq -2$, $x_1 \geq 0$, x_2 free.

Self-test 2. The primal is unbounded. What can the dual be? The dual is infeasible. What can the primal be?

Self-test 3. For $\max 3x_1 + 2x_2$ s.t. $x_1 + x_2 \le 4$, $x_1 + 3x_2 \le 9$, $x_1 \le 3$, $x \ge 0$, use complementary slackness to verify that $x = (3, 1)$ is optimal and find the dual solution.

Self-test 4. Prove Gordan's theorem: exactly one of "$\exists x : Ax < 0$" and "$\exists p \ge 0,\ p \ne 0,\ A^{\mathsf{T}}p = 0$" holds.

Self-test 5. A system $Ax = b$, $x \ge 0$ with 50 equations and 1000 unknowns is feasible. Show it has a solution with at most 50 nonzero entries.

Self-test 6. Forty half-planes in $\mathbb{R}^2$ are such that every three have a common point. Must all forty have a common point?

Self-test 7. True or false: if x and p are feasible for a primal–dual pair and $x_j > 0$ for some j with $p^{\mathsf{T}}A_j < c_j$, then x is not optimal.

Self-test 8. In HW3 P3, suppose the 500 inequalities are in 600 variables. What does the theorem give?

Answers

1. Max problem: natural direction is $\le$. Row 1 ($\le$): $y_1 \ge 0$; row 2 ($\ge$): $y_2 \le 0$. $x_1 \ge 0$ gives a natural ($\ge$) dual constraint, x_2 free gives an equality: $\min 4y_1 - 2y_2$ s.t. $y_1 + y_2 \ge 2$, $y_1 - y_2 = 1$, $y_1 \ge 0$, $y_2 \le 0$.

2. Dual must be infeasible. Primal is infeasible *or* unbounded (it cannot have a finite optimum).

3. Dual: $\min 4y_1 + 9y_2 + 3y_3$ s.t. $y_1 + y_2 + y_3 \ge 3$, $y_1 + 3y_2 \ge 2$, $y \ge 0$. At $x = (3, 1)$ row 2 is slack ($6 < 9$), so $y_2 = 0$; $x_1, x_2 > 0$ force $y_1 + y_3 = 3$ and $y_1 = 2$, so $y = (2, 0, 1) \ge 0$. Values: $11 = 8 + 0 + 3$. Optimal.

4. Not both: $0 = (A^{\mathsf{T}}p)^{\mathsf{T}}x = p^{\mathsf{T}}(Ax) < 0$ since $p \ge 0$, $p \ne 0$ and $Ax < 0$. At least one: $\max s$ s.t. $Ax + s\mathbf{1} \le 0$, $s \le 1$ is feasible ($x = 0, s = 0$) and bounded. Its dual is $\min q$ s.t. $A^{\mathsf{T}}p = 0$, $\mathbf{1}^{\mathsf{T}}p + q = 1$, $p, q \ge 0$. If no x has $Ax < 0$ the primal optimum is 0, so some dual-feasible point has $q = 0$, i.e. $\mathbf{1}^{\mathsf{T}}p = 1$: a nonzero $p \ge 0$ with $A^{\mathsf{T}}p = 0$.

5. The feasible set is a nonempty standard-form polyhedron with 50 equality rows; it has a BFS, which has at most 50 positive entries.

6. Yes: Helly's theorem with $n = 2$, $n + 1 = 3$.

7. False. The pair violates complementary slackness, so x and p are not *both* optimal, but the culprit may be p: any non-optimal dual-feasible p has such a slack against an optimal x. To prove x suboptimal you must show that *no* dual-feasible p is complementary to it.

8. An infeasible subsystem of at most $\min\{500, 601\} = 500$ inequalities: nothing. The result is only informative when $m > n + 1$.

8 One-page cheat sheet

Dual of $\min c^\mathsf{T} x$	$\max p^\mathsf{T} b$; rows $\geq / \leq / = \to p_i \geq 0 / \leq 0$/free; vars $\geq 0 / \leq 0$/free $\to p^\mathsf{T} A_j \leq / \geq / = c_j$
Lagrangian view	$g(p) = p^\mathsf{T} b + \min_x (c - A^\mathsf{T} p)^\mathsf{T} x$; signs of p make it a lower bound, dual constraints make it finite
Weak duality	$p^\mathsf{T} b \leq c^\mathsf{T} x$. Unbounded primal $\Rightarrow$ infeasible dual
Strong duality	optimal solution on one side $\Rightarrow$ optimal on the other, equal values; no gap for LPs
Impossible combinations	finite/unbounded, finite/infeasible, unbounded/unbounded
Both infeasible	possible
Complementary slackness	$p_i(a_i^\mathsf{T} x - b_i) = 0$, $(c_j - p^\mathsf{T} A_j) x_j = 0 \iff$ feasible pair is optimal
Geometry of optimality	$c \in \text{cone}\{a_i : i \text{ active at } x^\star\}$
Farkas	$Ax = b, x \geq 0$ XOR $p^\mathsf{T} A \geq 0,\ p^\mathsf{T} b < 0$
Farkas (inequalities)	$Ax \leq b$ XOR $p \geq 0,\ A^\mathsf{T} p = 0,\ b^\mathsf{T} p < 0$
Gordan	$Ax < 0$ XOR $p \geq 0,\ p \neq 0,\ A^\mathsf{T} p = 0$
HW3 P2	$(Ax \geq 0, x \geq 0 \Rightarrow x_1 = 0) \iff \exists p \geq 0 :\ p^\mathsf{T} A \leq 0,\ p^\mathsf{T} A_1 < 0$
Proving alternatives	easy direction: multiply and add; hard direction: LP + dual + strong duality
Strict inequality trick	homogeneous system: $p^\mathsf{T} A_1 < 0 \iff$ (after scaling) $p^\mathsf{T} A_1 \leq -1$
BFS sparsity	$\{Mx = d, x \geq 0\}$ nonempty $\Rightarrow$ solution with $\leq \text{rank}\, M$ nonzeros; optimal BFS exists if an optimum exists
HW3 P3	infeasible $Ax \leq b$ in $\mathbb{R}^n \Rightarrow$ infeasible subsystem of $\leq n+1$ rows (support of a dual vertex); tight
Helly / Carathéodory	every $n+1$ intersect $\Rightarrow$ all intersect; hull points need $\leq n+1$ generators
Phase-1 LP	$\min t$ s.t. $a_i^\mathsf{T} x - t \leq b_i;\ t^\star > 0 \iff$ infeasible; dual optimum is the certificate

Further reading. Bertsimas & Tsitsiklis, *Introduction to Linear Optimization*, Ch. 2 (BFS, §2.2–2.6) and Ch. 4 (duality, §4.1–4.7; Farkas in §4.6). Boyd & Vandenberghe, *Convex Optimization*, §5.1–5.2 (Lagrange dual), §5.5 (complementary slackness/KKT), §5.8 (theorems of alternatives).